Network Tariffs in GB

This page covers the main network tariffs in GB and how to use them in Gridcog

This section explains the two major network charges you'll see in the UK electricity market: **Distribution Use of System charges (DUoS)** and **Transmission Network Use of System charges (TNUoS)**. These are non-commodity (non-energy) costs that help recover the cost of building, operating and maintaining the distribution and transmission electricity networks respectively.

Summary Table

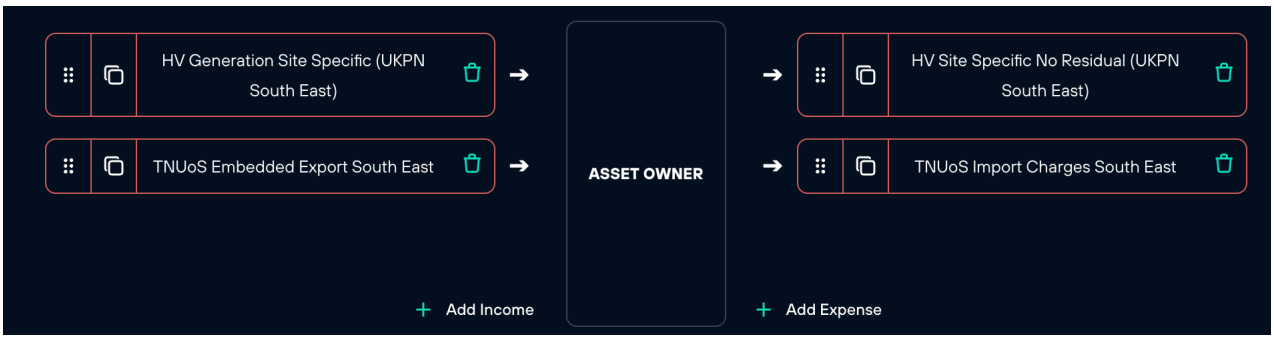
The table below provides an indicative view of which tariffs apply to different site types. Note that rules may change over time, so it's recommended to verify against actual bills where possible.

More information on each of these tariffs is provided below.

1. Pure final-demand import site distribution-connected (no export)	• **LV/HV Site Specific – Band X**	• **TNUoS Residual** • **TNUoS Import (Triads)**	e.g. commercial building, charge hub
2. Pure final-demand import site transmission-connected (no export)	• N/A	• **TNUoS Residual (T-Demand)** • **TNUoS Import (Triads)**	e.g. data centre, large industrial loads
3. Stand-alone storage / non-final demand site distribution-connected	• **LV/HV No Residual** for imports • **LV/HV Generation Site Specific** for exports	• **TNUoS Import (Triads)** for imports [in reality avoidable for BESS] • **TNUoS Embedded Export** for exports	e.g. distribution BESS / energy storage
4. Stand-alone storage / non-final demand site transmission-connected	• N/A	• **TNUoS Import (Triads)** for imports [in reality avoidable for BESS] • **Wider Generation TNUoS**	e.g. transmission BESS, pumped storage
5. Distribution-connected generator only (<100 MW, no demand)	• **LV/HV Generation Site Specific**	• **TNUoS Embedded Export (Triads)**	e.g. solar farm
6. Transmission-connected generator / Distribution-Connected TEC >100MW	• N/A	• **TNUoS Wider Generation**	e.g. onshore wind

7. Mixed final-demand + generation site Distribution Connected (BtM co-loc + generation) <100MW TEC	• **LV/HV Site Specific – Band X** for imports • **LV/HV Generation Site Specific** for exports	• **TNUoS Residual** • **TNUoS Import (Triads)** • **TNUoS Embedded Export (Triads)**	e.g. industrial load + generation + BESS
8. Mixed final-demand + generation site Transmission Connected (BtM co-location)	• N/A	• **TNUoS Residual (T-Demand)** • **TNUoS Import (Triads)** • **TNUoS Wider Generation**	e.g. data centre + generation + BESS
9. Co-located Renewables + BESS Import - Distribution	• **LV/HV No Residual** for imports • **LV/HV Generation Site Specific** for exports	• **TNUoS Import (Triads)** for imports (in reality avoidable for BESS) • **TNUoS Embedded Export** for exports	e.g. solar + BESS
10. Co-located Renewables + BESS Import - Transmission	• N/A	• **TNUoS Import (Triads)** for imports (in reality avoidable for BESS) • **TNUoS Wider Generation**	e.g. large renewables + BESS

Note that any tariffs that are meant for exports (Generation Site Specific and Embedded Export) are to be placed on the left hand side of the commercial model, as income. This matches the + and - signs used while building the item in the library.



Example Commercial Model Setup for a Standalone Battery in UKPN South East

For other complex physical site setups with non-final demand this guidance from NESO ↗ is a useful resource and here is additional information on co-located setups ↗. Note these network charges can be applied at the asset-level via submeters if you have separate BMUs for example onsite.

DUoS (Distribution Use of System)

Distribution Use of System (DUoS) are applied by your Distribution Network Operator (DNO) and vary by location, voltage level, time of use, and site type.

To find the right DNO for your postcode use the DNO Find my Network Operator ↗. This will inform which DUoS and TNUoS charges your site must pay.

LV/HV Site Specific – Band X

Applies to sites that consume energy for end-use (final demand).

Includes:

- Energy charge (£/kWh)
  - Varies by Red / Amber / Green time bands
- Fixed charge (p/day)
- Capacity charge (p/kVA/day)
- Reactive power charge (p/kVArh)

Voltage notes:

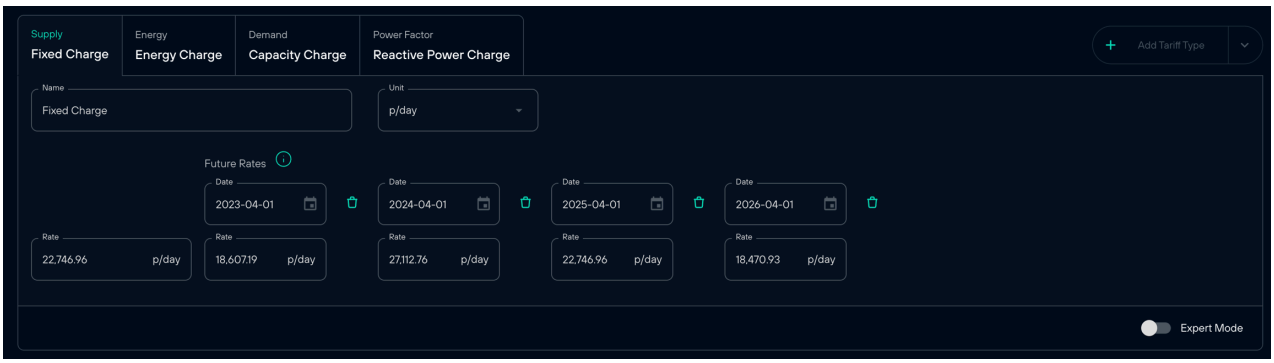
- LV → sites connected at low voltage
- HV → sites connected at 11–33 kV (depending on DNO)

Band notes: *(expand)*

The sites' band will depend on its Voltage connection, and on the Max Import Connection (MIC)

For small sites that don't have an MIC, the band is based on energy usage (kWh).

Unit	Band	Lower (>)	Upper (<=)	Lower (>)	Upper
		2021/22 to 2025/26	*2021/22 to 2025/26*	*2026/27 to 2030/31*	*2026/27 to*
LV no MIC (kWh)	Band 1	-	3,571	-	3,986
	Band 2	>3,571	12,553	>3,986	13,677
	Band 3	>12,553	25,279	>13,677	27,543
	Band 4	>25,279	∞	>27,543	∞
LV MIC (kVA)	Band 1	-	80	-	90
	Band 2	>80	150	>90	150
	Band 3	>150	231	>150	250
	Band 4	>231	∞	>250	∞
HV (kVA)	Band 1	-	422	-	500
	Band 2	>422	1,000	>500	1,100
	Band 3	>1,000	1,800	>1,100	2,000
	Band 4	>1,800	∞	>2,000	∞
EHV (kVA)	Band 1	-	5,000	-	3,500
	Band 2	>5,000	12,000	>3,500	11,000
	Band 3	>12,000	21,500	>11,000	20,000
	Band 4	>21,500	∞	>20,000	∞



Example ↗ of HV Site Specific tariff in the library

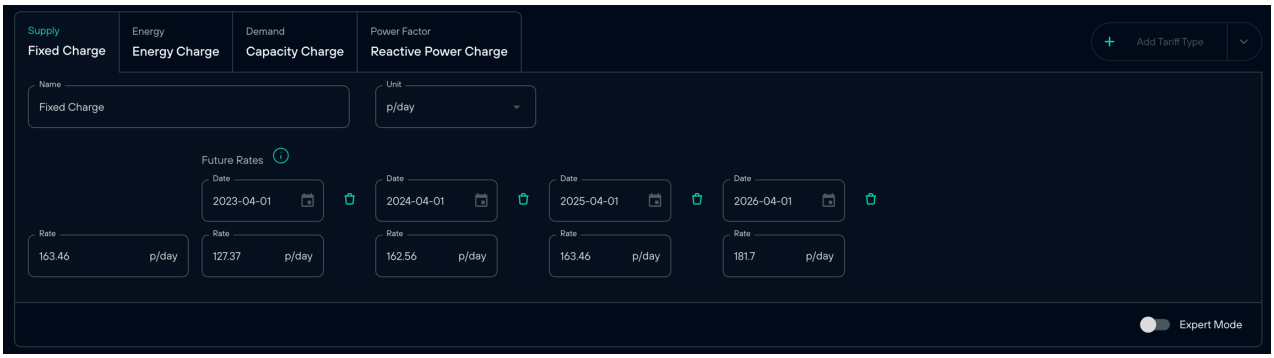
LV/HV Site Specific – No Residual

For non-final-demand sites such as battery storage (not consuming energy for own use).

Includes:

- Same structure as Final Demand tariffs
- Lower residual component, meaning:
  - Lower fixed charges
  - Lower unit (energy) charges in many cases

Used to avoid charging storage as if it were consumption.



Example ↗ of HV Site Specific No Residual tariff in the library

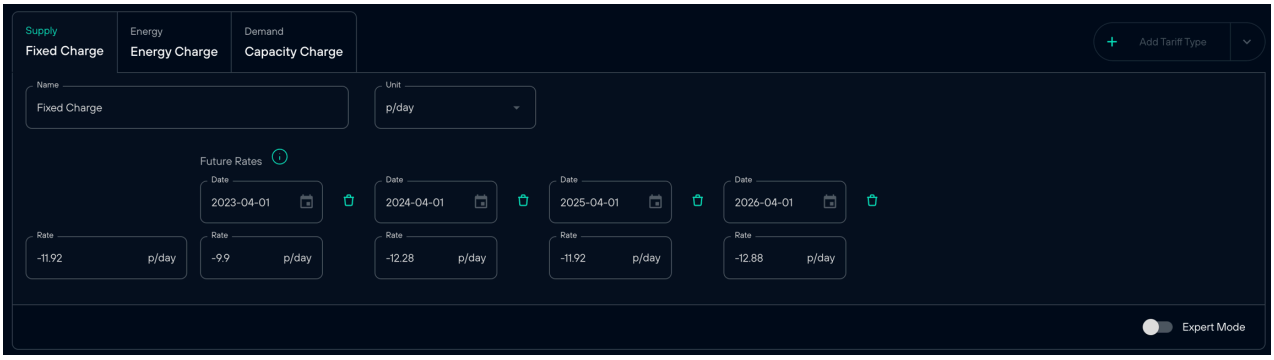
LV/HV Generation Site Specific (Exporting Sites)

For distribution-connected generation that exports (solar, wind, batteries exporting).

Includes:

- Export energy charge (£/kWh)
  - Often negative → a payment to the generator
  - Still uses Red/Amber/Green bands
- Fixed charge (p/day)

Reflects a generator's impact on local distribution constraints.



Example ↗ of HV Generation Site Specific tariff in the library

EHV Sites

Extra High Voltage sites will not be eligible for any of the above tariffs and will have a custom tariff agreed with the DNO. Using the site's MPAN number, these rates can be found in the Schedules of Charges documents published by the DNOs.

TNUoS (Transmission Network Use of System)

TNUoS are applied by the national electricity system operator (NESO) and recover the cost of building and maintaining the high-voltage transmission network. They vary by location (charging zone), whether the site is demand or generation, capacity/size of connection, and the charging year, and are typically levied on suppliers and passed through to end users.

Most TNUoS Charges are dependant on the network zone, whilst the wider generation tariffs are based upon the generation zone which is different to the DNO region.

For quick guide to TNUoS Charges see this handy NESO guide ↗.

Non-Locational Demand Residual Charge

In Gridcog: TNUoS Residual Charges
Applies to sites that consume energy for end-use (final demand).

Includes:

- Fixed daily charge (p/day)
  Based on voltage level and band.
  No time-of-day or seasonal variation.

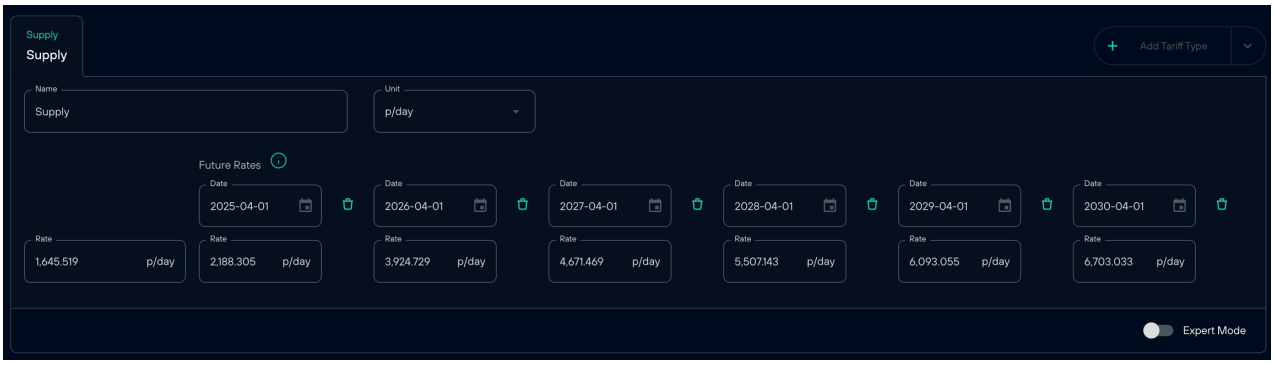
Band notes : ***(expand)***

The sites' band will depend on its Voltage connection, and on the Max Import Connection (MIC)

For small sites that don't have an MIC, the band is based on energy usage (kWh).

Unit	Band	Lower (>)	Upper (<=)	Lower$_2$ (>)	Upper$_2$
		2021/22 to 2025/26	*2021/22 to 2025/26*	*2026/27 to 2030/31*	*2026/27 to*
LV no MIC (kWh)	Band 1	-	3,571	-	3,986
	Band 2	>3,571	12,553	>3,986	13,677
	Band 3	>12,553	25,279	>13,677	27,543
	Band 4	>25,279	∞	>27,543	∞
LV MIC (kVA)	Band 1	-	80	-	90
	Band 2	>80	150	>90	150
	Band 3	>150	231	>150	250
	Band 4	>231	∞	>250	∞
HV (kVA)	Band 1	-	422	-	500
	Band 2	>422	1,000	>500	1,100
	Band 3	>1,000	1,800	>1,100	2,000
	Band 4	>1,800	∞	>2,000	∞
EHV (kVA)	Band 1	-	5,000	-	3,500
	Band 2	>5,000	12,000	>3,500	11,000
	Band 3	>12,000	21,500	>11,000	20,000
	Band 4	>21,500	∞	>20,000	∞
TCS (MWh)	Band 1	-	33,548	-	25131
	Band 2	33,548	73,936	25,131	64,451
	Band 3	73,936	189,873	64,451	163,88
	Band 4	189,873	∞	163,880	∞

Transmission connected sites will need the T-Demand residual charge



Example ↗ of TNUoS Residual tariff in the library

Half-Hourly Demand Charges (Import During Triads)

HH customers are charged according to the demand (MW) they take over the three Triad periods each year; the charge is levied through a £/kW tariff.

In Gridcog: TNUoS Import Charges
Applies to importing sites that are subject to Triads.

Includes:

- Charge based on import during Triads (p/kW)

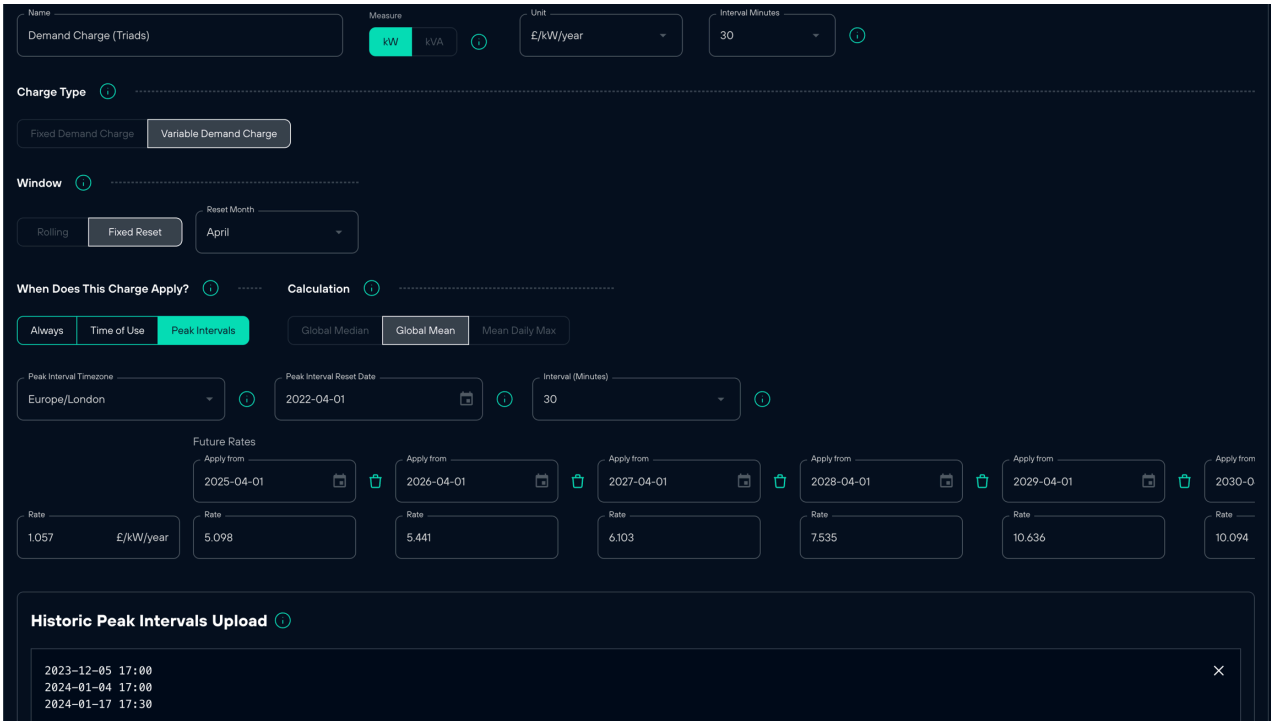
Triads are defined as:

- 3 × half-hour system peak periods
- between November and February (inclusive)
- Separated by ≥10 days

Only applies where the site's supplier and classification make it liable for Triad-based TNUoS charges.

Note that all batteries (distribution or transmission) are exposed to Half-Hourly Demand Charges charges if they import during a Triad. In reality only final-demand sites will import during triads.

Note that Northern Scotland, Southern Scotland, Northern, North West, Yorkshire, N Wales & Mersey do not have Half-Hourly Demand Charges.



Example ↗ of TNUoS Import tariff in the library

Embedded Export (Triad-Based Export Payments)

In Gridcog: TNUoS Embedded Export

Embedded Export Tariffs are paid by NESO to suppliers and embedded generators for distribution-connected generators exporting to the grid. In most cases, NESO pays the supplier who
then pays the embedded generator.
Your export volume is the average of your generation output over the Triad periods

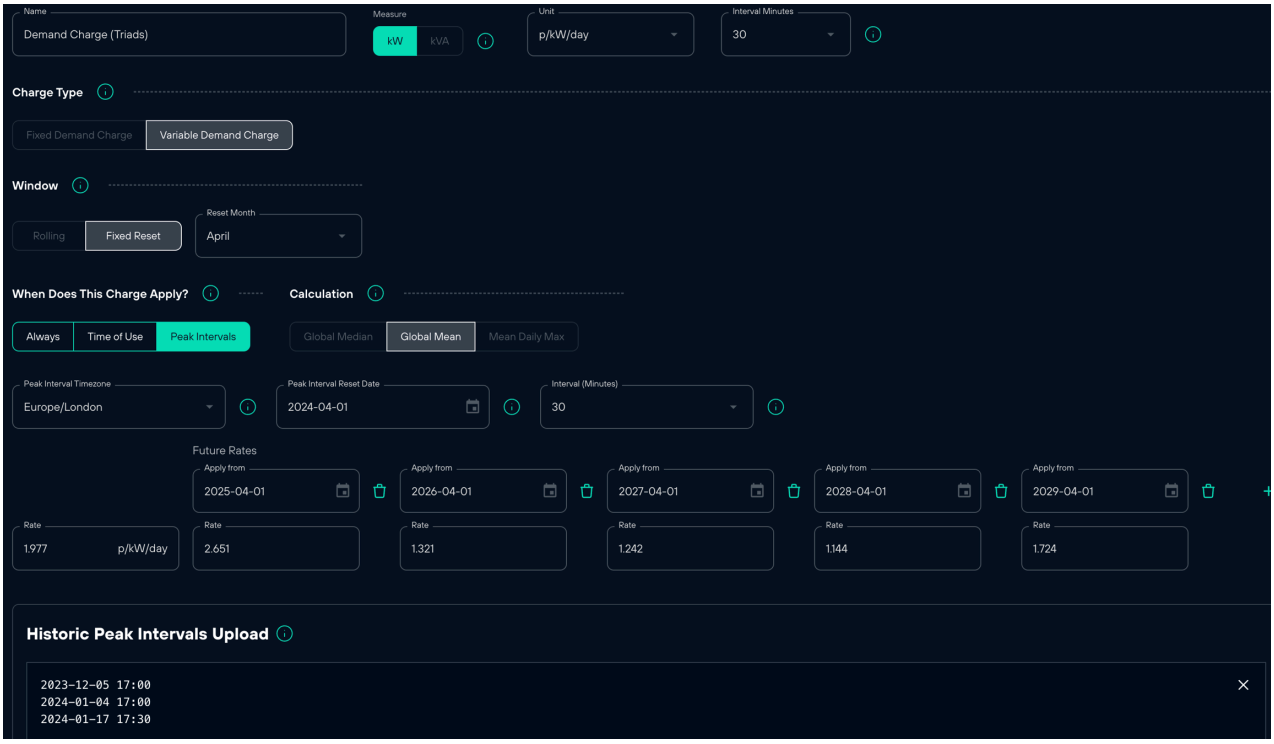
Our standard library items uses Embedded Export tariffs uses ESO's five-year regional Transmission Network Use of System (TNUoS) charges ↗. We assume that they remain flat from year 6 onwards.

Note that Northern Scotland, Southern Scotland, Northern, North West do not have EETs.

Includes:

- Payment (or charge) based on export during Triads

Applies only to exporting sites eligible for embedded benefits.



Example ↗ of TNUoS Embedded Export tariff in the library

Wider Generation Tariffs

Not a standard library item - specific per generator guidance is provided below to setup for your specific site

The Wider Generation Tariff is the main TNUoS charge applied to transmission-connected generators (and embedded generators ≥100 MW TEC). It reflects the generator's use of the whole transmission system rather than just local connection assets. Assets greater than 100MW and connected at distribution-level will only be subject to the wider generation tariffs.

Under today's methodology, each generation zone has four tariff elements, which combine differently depending on generator type. The wider tariff is made up of:

1. **Peak Security Tariff**
2. **Shared Year-Round Tariff**
3. **Not-Shared Year-Round Tariff**
4. **Adjustment Tariff**

Two of these may be multiplied by the generator's **Annual Load Factor (ALF)**, depending on whether the asset is intermittent or conventional.

The formula for the Wider Generation is as follows

Conventional Carbon Generators
(e.g. Biomass, CHP, Coal, Gas, Pumped Storage, Battery)

Peak Element + [Year Round Shared Element x ALF] + [Year Round Not Shared Element x ALF] + Adjustment Tariff

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Conventional Low Carbon Generators
(for example: Hydro, Nuclear)

Peak Element + [Year Round Shared Element x ALF] + Year Round Not Shared Element + Adjustment Tariff

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Intermittent Generators
(for example: Wind, Wave, Tidal, Solar)

[Year Round Shared Element x ALF] + Year Round Not Shared Element + Adjustment Tariff

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ALFs are used to scale the Shared Year-Round element of tariffs for each generator, and the Year Round Not Shared for Conventional Carbon generators, so that each has a tariff appropriate to its historical load factor.

e.g. based on these charges batteries connecting in at the transmission network in Scotland will face charges; in contrast assets in Scotland connecting in at distribution level will avoid these. Batteries in the South West by contrast will see net benefit from these Wider Generation Tariffs

Annual Load Factor By Technology ***(expand)***

Annual Load Factor from NESO↗'s Five-Year View of TNUoS Tariffs for 2026/27 – 2030/31

Technology	Generic ALF (%)
Battery	3.8884%
Biomass	42.9869%
CCGT_CHP	42.3027%
Coal	29.0586%
Gas_Oil	0.8252%
Hydro	39.6894%
Nuclear	55.6863%
Offshore_Wind	48.2176%
Onshore_Wind	41.5111%
Pumped_Storage	9.4949%
Reactive_Compensation	0.0000%
Solar	10.8000%
Tidal	13.2000%
Wave	2.9000%

Wider Generation Tariff per Generation Region ***(expand)***

Generation Wider Tariffs in 2026/27

Zone Name	System Peak Tariff (£/kW)	Shared Year Round Tariff (£/kW)	Not Shared Year Round Tariff (£/kW)	Adjustment Tariff (£/kW)
North Scotland	3.064973	28.42961	16.38866	-2.23144
East Aberdeenshire	4.351353	17.23317	16.38866	-2.23144
Western Highlands	3.081449	28.22489	16.28487	-2.23144
Skye and Lochalsh	-4.96846	28.22489	15.88969	-2.23144
Eastern Grampian and Tayside	5.063745	21.80893	12.25195	-2.23144
Central Grampian	4.884967	22.19994	12.6439	-2.23144
Argyll	4.127851	20.20401	23.91296	-2.23144
The Trossachs	3.482386	20.20401	10.61215	-2.23144
Stirlingshire and Fife	2.852331	19.35567	10.07253	-2.23144
South West Scotlands	1.99015	18.54696	9.693199	-2.23144
Lothian and Borders	3.566901	18.54696	3.424582	-2.23144
Solway and Cheviot	1.876509	11.28206	5.883586	-2.23144
North East England	3.988013	7.551269	3.817741	-2.23144

North Lancashire and The Lakes	2.010215	7.551269	-0.08553	-2.23144
South Lancashire, Yorkshire and Humber	5.031594	2.342314	0.625945	-2.23144
North Midlands and North Wales	3.717393	0.689317		-2.23144
South Lincolnshire and North Norfolk	0.120916	2.591365		-2.23144
Mid Wales and The Midlands	0.538499	3.302727		-2.23144
Anglesey and Snowdonia	6.054	1.294987		-2.23144
Pembrokeshire	7.9832	-9.16886		-2.23144
South Wales & Gloucester	3.306105	10.03599		-2.23144
Cotswold	0.944809	4.2329	13.59098	-2.23144
Central London	-4.13023	4.2329	6.815348	-2.23144
Essex and Kent	-3.43458	4.2329		-2.23144
Oxfordshire, Surrey and Sussex	-0.94189	3.750954		-2.23144
Somerset and Wessex	-3.51988	4.519946		-2.23144

West Devon and Cornwall	-3.32375	13.76799		-2.23144

Generation Wider Tariffs in 2027/28

Zone Name	System Peak Tariff (£/kW)	Shared Year Round Tariff (£/kW)	Not Shared Year Round Tariff (£/kW)	Adjustment Tariff (£/kW)
North Scotland	3.5268	29.60564	19.34597	-2.61643
East Aberdeenshire	6.154921	15.79355	19.34597	-2.61643
Western Highlands	2.360197	26.92728	17.64588	-2.61643
Skye and Lochalsh	-5.81024	26.92728	17.24527	-2.61643
Eastern Grampian and Tayside	5.585642	21.99366	13.91445	-2.61643
Central Grampian	5.389964	21.84434	13.73884	-2.61643
Argyll	4.618902	19.97269	27.02374	-2.61643
The Trossachs	3.896022	19.97269	11.50853	-2.61643
Stirlingshire and Fife	3.542705	19.20066	10.95777	-2.61643
South West Scotlands	2.493736	18.6767	10.6739	-2.61643
Lothian and Borders	4.334974	18.6767	3.903921	-2.61643
Solway and Cheviot	2.334365	11.61207	6.610433	-2.61643
North East England	5.357886	7.55433	4.048902	-2.61643

North Lancashire and The Lakes	2.312481	7.55433	0.685179	-2.61643
South Lancashire, Yorkshire and Humber	5.782244	2.12717	0.272731	-2.61643
North Midlands and North Wales	3.865566	0.598605		-2.61643
South Lincolnshire and North Norfolk	0.722959	2.17839		-2.61643
Mid Wales and The Midlands	0.476606	2.974421		-2.61643
Anglesey and Snowdonia	4.665381	0.949615		-2.61643
Pembrokeshire	5.257229	-11.712		-2.61643
South Wales & Gloucester	1.441643	11.88742		-2.61643
Cotswold	0.709048	3.898146	13.57127	-2.61643
Central London	-4.27151	3.898146	6.908697	-2.61643
Essex and Kent	-3.60428	3.898146		-2.61643
Oxfordshire, Surrey and Sussex	-1.0455	3.810067		-2.61643
Somerset and Wessex	-3.45275	5.183991		-2.61643

West Devon and Cornwall	-3.93447	14.27653		-2.61643

Generation Wider Tariffs in 2028/29

Zone Name	System Peak Tariff (£/kW)	Shared Year Round Tariff (£/kW)	Not Shared Year Round Tariff (£/kW)	Adjustment Tariff (£/kW)
North Scotland	3.465234	38.38735	28.18825	-3.07674
East Aberdeenshire	4.867084	8.259951	27.93267	-3.07674
Western Highlands	3.839246	25.00147	18.00806	-3.07674
Skye and Lochalsh	3.748311	25.00147	25.90306	-3.07674
Eastern Grampian and Tayside	6.231035	21.12381	14.54547	-3.07674
Central Grampian	5.583498	21.13562	14.56083	-3.07674
Argyll	4.546294	19.27015	27.88275	-3.07674
The Trossachs	3.708178	19.27015	12.1047	-3.07674
Stirlingshire and Fife	4.282679	17.51387	10.68425	-3.07674
South West Scotlands	2.126473	18.16548	11.02331	-3.07674
Lothian and Borders	4.044085	18.16548	4.560002	-3.07674
Solway and Cheviot	2.038637	11.16973	6.386732	-3.07674
North East England	4.612641	7.780509	3.873933	-3.07674

North Lancashire and The Lakes	1.796319	7.780509	0.315571	-3.07674
South Lancashire, Yorkshire and Humber	5.739909	2.404097	0.351874	-3.07674
North Midlands and North Wales	4.033148	0.699557	0.015387	-3.07674
South Lincolnshire and North Norfolk	0.592599	1.882766	0.015387	-3.07674
Mid Wales and The Midlands	0.86685	2.649649	0.015387	-3.07674
Anglesey and Snowdon	4.593465	1.22873	0.015387	-3.07674
Pembrokeshire	8.864785	-6.3065		-3.07674
South Wales & Gloucester	3.947014	7.419301		-3.07674
Cotswold	2.608502	3.108723	11.17775	-3.07674
Central London	-4.54714	3.108723	6.738199	-3.07674
Essex and Kent	-3.99136	3.108723		-3.07674
Oxfordshire, Surrey and Sussex	-1.669	-4.62254		-3.07674
Somerset and Wessex	-4.38768	7.107035		-3.07674

West Devon and Cornwall	1.490447	13.26709		-3.07674

Generation Wider Tariffs in 2029/30

Zone Name	System Peak Tariff (£/kW)	Shared Year Round Tariff (£/kW)	Not Shared Year Round Tariff (£/kW)	Adjustment Tariff (£/kW)
North Scotland	4.957249	58.29414	37.33249	-5.86856
East Aberdeenshire	4.067684	33.68804	34.83045	-5.86856
Western Highlands	5.546874	46.2126	25.42834	-5.86856
Skye and Lochalsh	5.248789	46.2126	37.3323	-5.86856
Eastern Grampian and Tayside	5.106724	40.621	22.47073	-5.86856
Central Grampian	6.21859	38.71371	21.05688	-5.86856
Argyll	5.172259	39.19559	29.77887	-5.86856
The Trossachs	4.116551	35.57727	18.70575	-5.86856
Stirlingshire and Fife	3.918183	34.30115	18.03148	-5.86856
South West Scotlands	2.70514	30.64004	16.79298	-5.86856
Lothian and Borders	4.054377	30.64004	12.56086	-5.86856
Solway and Cheviot	1.676155	18.01671	10.43117	-5.86856
North East England	4.906533	10.15512	6.04719	-5.86856

North Lancashire and The Lakes	1.587337	10.15512	0.961055	-5.86856
South Lancashire, Yorkshire and Humber	5.541669	0.37019	0.304814	-5.86856
North Midlands and North Wales	3.773242	-0.97458	-0.0559	-5.86856
South Lincolnshire and North Norfolk	0.438823	0.576453	-0.02809	-5.86856
Mid Wales and The Midlands	0.679281	0.144132	0.001046	-5.86856
Anglesey and Snowdon	4.694089	0.693955	-0.0559	-5.86856
Pembrokeshire	9.348548	-9.32284		-5.86856
South Wales & Gloucester	4.290857	10.52282		-5.86856
Cotswold	2.948673	0.156206	11.01706	-5.86856
Central London	-4.47042	0.156206	6.611872	-5.86856
Essex and Kent	-3.90116	0.156206		-5.86856
Oxfordshire, Surrey and Sussex	-1.36563	-7.66707		-5.86856
Somerset and Wessex	-4.38562	-10.6965		-5.86856

West Devon and Cornwall	-1.27578	-16.3645		-5.86856

Generation Wider Tariffs in 2030/31

Zone Name	System Peak Tariff (£/kW)	Shared Year Round Tariff (£/kW)	Not Shared Year Round Tariff (£/kW)	Adjustment Tariff (£/kW)
North Scotland	8.696164	68.34327	25.54125	-4.8285
East Aberdeenshire	6.461708	45.99053	23.26834	-4.8285
Western Highlands	9.724084	53.43058	17.35345	-4.8285
Skye and Lochalsh	9.417649	53.43058	29.50084	-4.8285
Eastern Grampian and Tayside	8.360783	47.62209	15.32482	-4.8285
Central Grampian	7.026964	42.9389	13.11786	-4.8285
Argyll	9.867748	45.04332	27.13495	-4.8285
The Trossachs	8.338758	41.51931	12.46368	-4.8285
Stirlingshire and Fife	6.508404	40.44227	12.14993	-4.8285
South West Scotlands	4.382653	37.40071	11.62287	-4.8285
Lothian and Borders	3.612517	37.40071	6.607612	-4.8285
Solway and Cheviot	2.161957	22.13832	7.389419	-4.8285
North East England	4.864566	10.08454	3.573647	-4.8285

North Lancashire and The Lakes	1.951627	10.08454	0.870683	-4.8285
South Lancashire, Yorkshire and Humber	5.268582	0.290033	0.326401	-4.8285
North Midlands and North Wales	3.2858	-2.05522	0.117763	-4.8285
South Lincolnshire and North Norfolk	-0.86753	-0.2393	0.022226	-4.8285
Mid Wales and The Midlands	0.130328	0.256778	0.009944	-4.8285
Anglesey and Snowdon	5.425857	2.332786	0.117763	-4.8285
Pembrokeshire	8.625351	-10.0483		-4.8285
South Wales & Gloucester	3.433317	11.78687		-4.8285
Cotswold	3.32111	0.131618	11.09338	-4.8285
Central London	-5.29895	0.131618	6.517212	-4.8285
Essex and Kent	-3.83161	0.131618		-4.8285
Oxfordshire, Surrey and Sussex	-2.18883	-6.47407		-4.8285
Somerset and Wessex	1.614534	-7.83549		-4.8285

West Devon and Cornwall	2.045231	-12.0483		-4.8285

Custom Tariffs

- When creating custom tariffs as separate library items remember to assign imports as expenses, and exports as income within the Commercial Model assignment.
- If you add both export and import charges to the same library item and this item is assigned to the expense side in the commercial model the export charges should include a negative sign to ensure they are considered a "negative expense" [i.e. revenue].

The rates and information on how the tariff elements add together can be found in NESO's portal ↗